

Electricity Consumption Analysis in Tableau

Plugging into the Future: An Exploration of Electricity Consumption Patterns in India Using Tableau

Introduction

Electricity consumption analysis helps understand energy usage across states, regions, and time periods. In this project Tableau is used to convert raw electricity consumption data into interactive dashboards and stories that support better decision making. Tableau enables users to identify trends, compare regions, and analyze yearly, monthly, and quarterly consumption patterns.

Problem Statement

Raw electricity datasets are difficult to interpret using spreadsheets alone. This project addresses the problem by creating interactive Tableau visualizations that simplify analysis and highlight meaningful insights.

Objectives

- Analyze electricity consumption.
- Compare 2019 and 2020.
- Study regional and monthly trends.
- Build dashboards and stories.
- Publish results using Tableau Public.

Dataset Description

The dataset contains Date, Year, Month, Quarter, State, Region, and Consumption fields. The data was pre-cleaned, after which data types, field names, and values were verified before analysis.

Data Preparation

The dataset was imported into Tableau, field data types were validated, columns were renamed for readability, calculated fields were created where required, and totals were validated against the source dataset.

Visualizations

The project includes state-wise consumption for 2019 and 2020, total consumption, usage by region, total regional consumption, top and bottom consuming states, month-wise trends, quarter-wise trends, region-wise state usage, metro city analysis, yearly comparison, and before/after lockdown analysis.

Dashboard

Multiple worksheets were combined into a dashboard with filters for State, Region, Year, and Date. Users can interactively explore the data and compare consumption across dimensions.

Storyboard

The Tableau Story presents the project sequentially: introduction, yearly comparison, regional analysis, monthly trends, lockdown impact, dashboard summary, and key findings.

Results & Insights

The analysis shows considerable variation in electricity consumption across states and regions. Monthly trends indicate seasonal demand, while the lockdown period produced noticeable changes in consumption patterns. Interactive dashboards improve understanding and exploration.

Conclusion

The project demonstrates how Tableau transforms raw electricity consumption data into clear and interactive visualizations. The dashboards and story make it easier to analyze patterns and support informed decision-making.

Future Scope

Future enhancements include adding recent years, real-time data, predictive analytics,